# Gabriele Durante

*Junior Data/ML Engineer — EU Vulcanus in Japan Fellow '25/'26*

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## Professional Summary

Completing MSc in **Data Science** (graduating July 2026) with hands-on internship experience in production ML systems (LLM agents, RAG) and applied research in **NLP/CV**. Recently completed the Vulcanus in Japan fellowship in Tokyo specializing in **real-time Transformer optimization**. Seeking an entry-level ML Engineer role post-graduation to apply deep learning expertise in production environments.

## Technical Skills

- **Data & Infrastructure**: Python, SQL, Docker, Git/GitHub Actions, FastAPI, CI/CD Pipelines, Linux Environment.
- **ML Engineering**: PyTorch, Transformers (Hugging Face), ONNX, RAG, LangChain, NLP/NER, Computer Vision (YOLO, OpenCV).
- **Training & Optimization**: DDP, AMP (fp16), Weights & Biases, Signal Processing, Time Series Analysis, Statistical Modeling.

## Work Experience

- **Equmenopolis, Inc.** — **Tokyo, Japan**
  *ML Research Intern (Vulcanus Fellow)* — *Aug 2025 – May 2026*
  - Authored **SynapseLink**, a 13.1M-parameter Transformer for audio-to-blendshape generation. Achieved **204× real-time** inference, reducing compute requirements by **85%** and enabling cost-effective deployment on edge devices.
  - Designed **AdaptiveArticulationLoss** to deterministically prevent mean-face collapse, achieving a **+91.8% expressiveness improvement** and resulting in highly realistic AI avatars for enhanced user engagement.
  - Optimized training pipeline with **fp16 AMP mixed-precision** and a pre-computed tensor cache, reducing I/O bottlenecks by **57×** and cutting cloud compute costs by **25%**.
  - Engineered a production-ready streaming inference pipeline for **Unity/Unreal Engine integration**, reducing deployment time for interactive experiences from 2 months to **3 weeks**.
- **HPA (High Performance Analytics)** — **Trento, Italy**
  *Data Scientist Intern (Full-time transition)* — *Feb 2025 – July 2025*
  - Built a privacy-focused **LLM agent system** leveraging custom SearXNG. Designed a RAG architecture pairing reasoning LLMs with vector stores, reducing retrieval latency from 1.2s to **350ms**.
  - Developed **BonsaiParser**, a modular document framework for RAG ingestion; engineered LLM action pipelines transforming checklists into automated workflows with **100% observability** on agent reasoning steps.
  - Conducted applied research on Computer Vision (WASTE model): fine-tuned **YOLO** architectures to improve object detection accuracy; delivered full test coverage and CI/CD deployment via Docker and GitHub Actions.

## Projects & Publications

- **SynapseLink (Publication)**: Released the production-ready Transformer model developed during the Vulcanus fellowship. Achieved **204× real-time** inference and **+91.8% expressiveness** for audio-to-avatar generation. Optimized for low-latency edge deployment.
- **NerGuard: Scalable Multilingual PII Detection System (Master's Thesis)**: Fine-tuned **mDeBERTa-v3** on 580K entries across 8 languages with DDP training and entropy-based span-level LLM routing. Achieved **span F1 = 0.904** (+186% vs. Presidio, +102% vs. GLiNER), reduced LLM calls by **50%**, and reached **∼25ms CUDA latency**. Open source.
- **tensionr**: high-fidelity intelligence open-source dashboard designed to monitor global instability, narrative shifts, and strategic asset movements in real-time by synthesizing multi-domain signals (GDELT news flows, ADS-B aerial telemetry) providing unified geopolitical tactical picture for rapid situational awareness.
- **Deep Learning for Venusian Volcanic Detection (Magellan SAR Data)**: **Developed** CNN architectures for noisy SAR imagery using SMOTE and PCA preprocessing. Achieved **95.31% test accuracy (0.97 AUC)**, outperforming Random Forest and FFNN baselines.

## Education

- **University of Verona** — **Verona, Italy**
  *MSc in Data Science – Focus: Deep Learning, NLP, Text Mining* — *Graduation: July 2026*
- **University of Oslo** — **Oslo, Norway**
  *MSc in Computer Science, Exchange Program – Algorithms, Deep Reinforcement Learning* — *2024 – 2025*
- **University of Brescia** — **Brescia, Italy**
  *BSc in Banking and Finance – Focus: Econometrics, Statistics, Financial Modeling* — *Oct 2023*

## Honors & Awards

- **Vulcanus in Japan Programme – EU-Japan Centre for Industrial Cooperation** — **Tokyo, Japan**
  *Selected as **1 of 23 fellows** from 500+ STEM candidates; co-funded by the European Commission and METI.Aug 2025 – May 2026*